DATASCI 224 Final Project Report
Classifying Pediatric Chest X-Rays with a Vision Transformer

Section 1: Describing the Problem and the Dataset

Pneumonia is an acute lung infection and one of the leading infectious causes of death among young children worldwide [1]. The chest X-ray (CXR) is the most common first-line test for suspected pneumonia, but reading one accurately takes a trained radiologist, different readers do not always agree, and results can be delayed where specialists are scarce [2]. An automated classifier could help as a triage tool, flagging the X-rays most likely to show pneumonia for faster review. This project examines whether a fine-tuned Vision Transformer (ViT) can classify pediatric chest X-rays as normal or pneumonia. Convolutional Neural Networks (CNNs) are the usual choice for medical image classification, but I chose a ViT to test a newer, attention-based approach on a clinically important task.

Dataset: I used the publicly available Chest X-Ray Images (Pneumonia) dataset from Kermany et al. (2018) [2], obtained from Kaggle. The images come from children aged 1 to 5, taken during routine care at the Guangzhou Women and Children's Medical Center in China, and were screened for quality and graded by expert physicians. The dataset has about 5,856 grayscale images of varying resolution, already split into a training set of 5,216 images (1,341 normal; 3,875 pneumonia) and a test set of 624 images (234 normal; 390 pneumonia). The task is to label each image as normal or pneumonia.

Two features of the data shaped my approach. First, the classes were imbalanced: pneumonia makes up about 74% of the training images, so accuracy alone can overstate how well a model is doing. I therefore also report AUC, precision, recall, and F1. Second, the official validation set had only 16 images (eight per class), too few to track training reliably. I set it aside and instead held out a class-balanced 15% of the training images for validation, keeping the 624 test images for one final evaluation.

Before modeling, each grayscale X-ray was copied into three channels, resized to 224×224 pixels, and scaled to the value range the model saw during pretraining. I applied light augmentation (small rotations and shifts) to the training images only, and did not flip images left to right, since the chest is not symmetrical.

Figure 1 shows sample X-rays and Figure 2 shows the class counts.

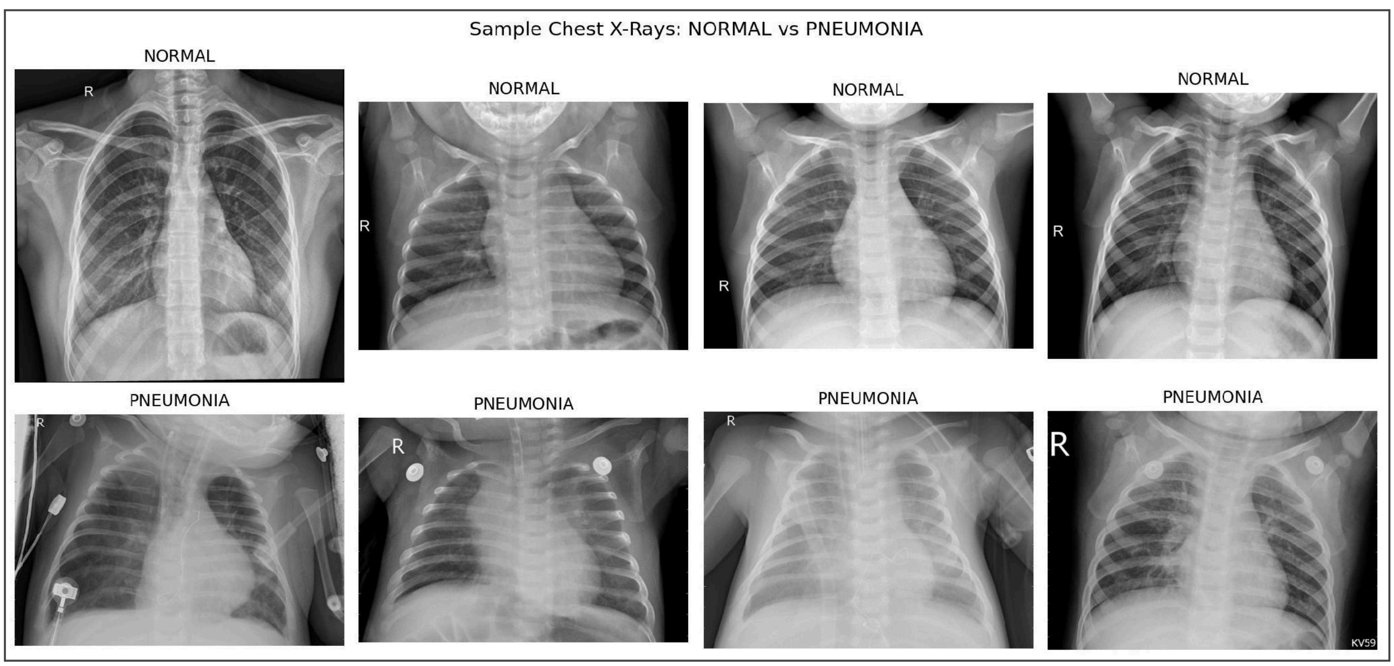

Figure 1. Representative chest X-rays: normal cases (top row) and pneumonia cases (bottom row). Pneumonia films typically show increased opacity in the lung fields.

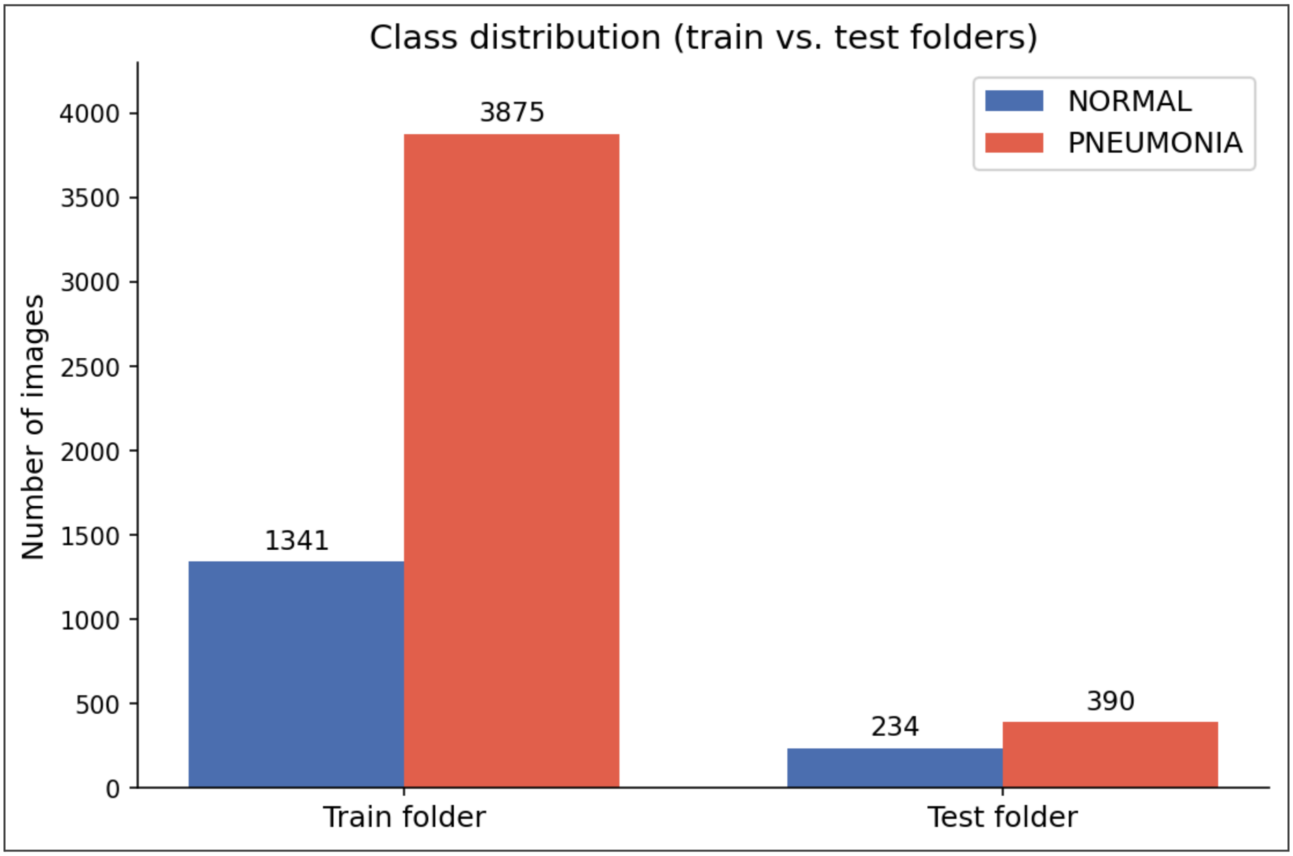

Figure 2. Class distribution across the training and test folders.

Section 2: Describing the Vision Transformer

The Vision Transformer (ViT), introduced by Dosovitskiy et al. (2020) [3], applies the transformer design built for language [4] to images instead. A convolutional neural network (CNN) builds an image up from small local regions, but a ViT uses attention, so every part of the image can compare itself to every other part at once. This whole-image view suits pneumonia well, since its main sign, increased opacity (cloudy areas), can appear anywhere in the lungs and cover a large region rather than a single edge. A ViT is also practical here: strong pretrained versions are free to download, so a small medical dataset can reuse what a model already learned from 14 million images.

How it Works: A ViT works in three main steps: it splits the image into patches, compares the patches using attention, and combines them into a single prediction. First, the image is split into patches. An image of height H and width W with C color channels is cut into N square patches of side P, where $N = HW / P^2$. A 224×224 image with P = 16 gives N = 196 patches. Each patch is flattened (16×16×3 = 768 numbers) and passed through a learnable linear layer E to become a vector of length D (D = 768 for ViT-Base). This vector is the patch's embedding, the image version of turning a word into numbers in a language model. A special classification token is added to the front, and a learned position tag (E_pos) is added so the model knows where each patch came from.

This sequence passes through L = 12 identical encoder blocks. Each block runs multi-head self-attention (MSA) and a small feed-forward network (MLP). Each step is layer-normalized (LN) first, then wrapped in a residual connection that adds the input back to the output so information is not lost as it moves through the layers.

The key step is attention. Each patch produces three vectors: a query Q (what this patch is looking for), a key K (what information it contains), and a value V (what information it passes on). A patch's output is a weighted mix of all the values, where each weight depends on how well one patch's key matches another's query:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(Q K^T / \sqrt{d} \right) V$$

Here $Q K^T$ measures how similar two patches are, dividing by $\sqrt{d}$ keeps the numbers stable, and the softmax turns them into weights that add up to one. The patches the model finds most relevant get the most weight in producing each patch's output. Using several attention "heads" lets the model look for different relationships at once. After the last block, the model takes the classification token's final vector and feeds it to a small MLP that outputs the two class scores. In short, the math is linear algebra (the patch and Q/K/V projections), the dot product for similarity, the softmax, residual connections and normalization for stable training, and cross-entropy loss. ViT-Base/16 has about 86 million parameters; fine-tuning replaces its pretrained classification head with a fresh two-class one and retrains with a small learning rate.

A ViT does make some assumptions. It treats an image as a sequence of fixed-size patches and relies on attention plus learned position information to relate them. Because it has few built-in assumptions about images (unlike a CNN, which assumes that nearby pixels belong together), it needs a lot of data: trained from scratch on a small dataset it usually loses to a CNN, and it only catches up after being pretrained on a large dataset and then fine-tuned [3]. That is exactly the setup here. The one catch is that features learned from everyday photos must still be useful on grayscale X-rays; the two are different, but fine-tuning is what closes that gap.

Section 3: Architecture, Parameter Selection, and Discussion of Results

Experimental Setup: The google/vit-base-patch16-224-in21k model from the HuggingFace library was fine-tuned using the stratified 85/15 training/validation split described in Section 1, with the 624-image test set held out for final evaluation. To address the class imbalance, the cross-entropy loss was weighted by inverse class frequency, thereby increasing the penalty for errors on the minority (normal) class. Optimization employed the AdamW optimizer (a standard choice for fine-tuning transformers) with a learning rate of 2×10^{-5}, a batch size of 16, and 5 epochs. Loss and AUC were monitored on the validation split at each epoch (Figure 4), and the final model was evaluated once on the held-out test set (Figure 3).

Results: The performance of the fine-tuned model on the held-out test set is reported in Figure 3. For context, the table additionally lists values reported in the published literature for ViT-family models evaluated on the same dataset; these constitute external reference points rather than results of the present analysis.

Model / Source	Acc.	AUC	Prec.	Recall	F1
My Work: fine-tuned ViT-Base/16 (test set)	**0.862**	**0.932**	**0.867**	**0.920**	**0.893**
Reference: DeiT-Base/16, Sci. Reports 2024 [5]	0.976	NA	NA	0.95	NA

Figure 3. Test-set metrics for the fine-tuned ViT (top) alongside published reference results on the same dataset. NA indicates a metric not reported by Singh et al. [5].

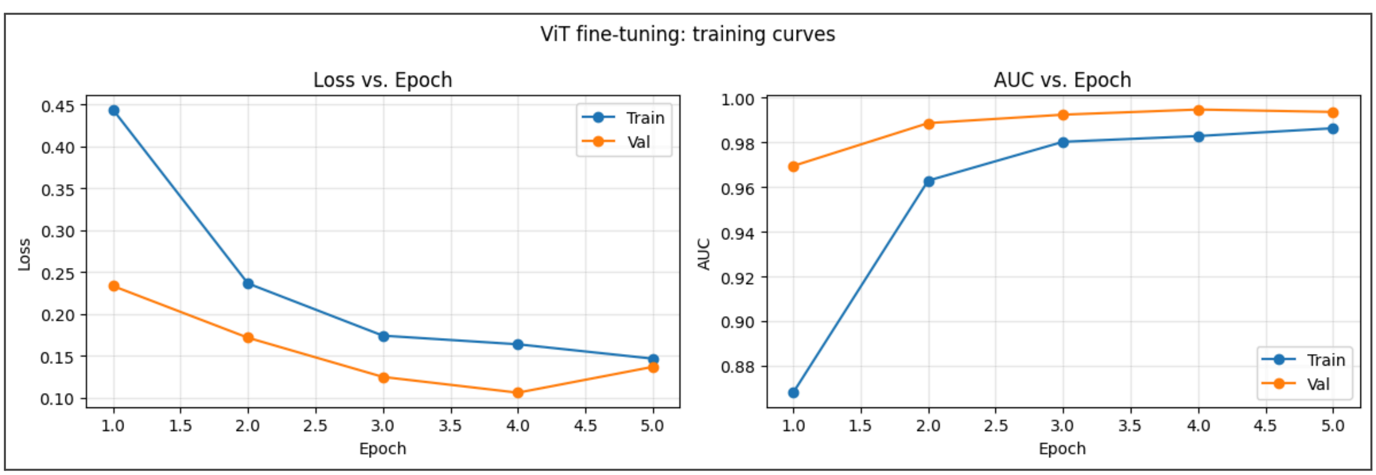

Figure 4. Loss and AUC versus epoch on the training and validation splits during fine-tuning.

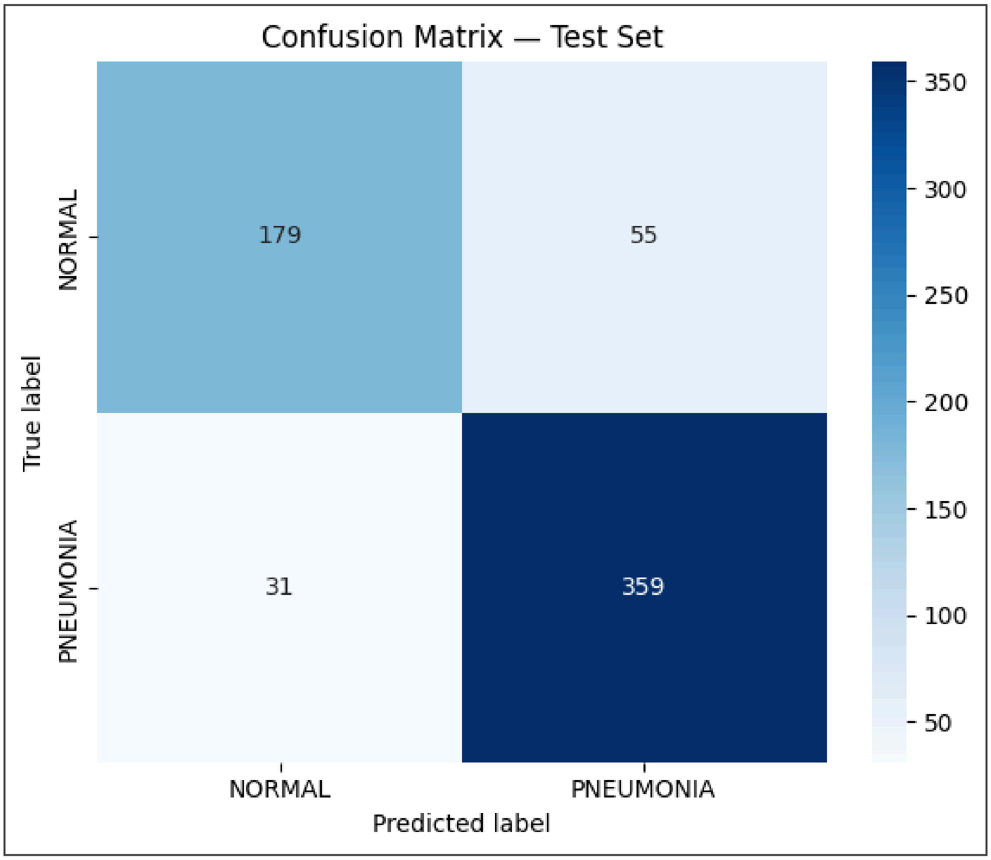

Figure 5. Confusion matrix on the held-out 624-image test set.

Discussion: The fine-tuned ViT reached 86.2% accuracy and an AUC of 0.932 on the test set. That is a little below the published result in Figure 3 (97.6%), which used heavier tuning and augmentation. The training curves in Figure 4 show clean learning: training loss dropped steadily from 0.44 to 0.15, and validation AUC rose to about 0.99 by epoch 4. Validation loss was lowest at epoch 4 (0.106) and rose a little at epoch 5 (0.137) while training loss kept falling, which is the first mild sign of overfitting; the weighted loss and light augmentation otherwise kept it in check.

Since pneumonia is the majority class, accuracy by itself is misleading. The number that matters most is recall on pneumonia, and here the model did well: 0.92 recall, meaning it caught 359 of the 390 pneumonia images and missed only 31 (Figure 5). The cost shows up on the normal class, with a lower recall of 0.76, since the model labeled 55 of 234 normal images as pneumonia. In other words, it makes more false alarms than missed cases. For a tool meant to flag images for review, that is the safer mistake, because sending a healthy X-ray for a second look matters far less than missing a real pneumonia. It is also the expected result of the class imbalance and the weighted loss.

A few limits are worth noting. All of the data comes from young children at one hospital, so the model may not do as well on adults or on images from other places. It also assumes that features learned from everyday photos carry over to grayscale X-rays; earlier work shows a ViT used straight out of the box does poorly on raw X-rays and that fine-tuning is what makes it work [6], which is why I fine-tuned it here. Finally, the official 16-image validation set was too small to be useful, so I made my own split. Overall, the results show that a pretrained Vision Transformer, fine-tuned with care for the class imbalance, is a strong and sensible choice for telling normal and pneumonia chest X-rays apart.

References

1. World Health Organization. Pneumonia in children. WHO Fact Sheets. Updated November 11, 2022. Accessed 2025. https://www.who.int/news-room/fact-sheets/detail/pneumonia

2. Kermany DS, Goldbaum M, Cai W, et al. Identifying medical diagnoses and treatable diseases by image-based deep learning. Cell. 2018;172(5):1122-1131.e9. doi:10.1016/j.cell.2018.02.010

3. Dosovitskiy A, Beyer L, Kolesnikov A, et al. An image is worth 16×16 words: transformers for image recognition at scale. In: International Conference on Learning Representations (ICLR); 2021. arXiv:2010.11929.

4. Vaswani A, Shazeer N, Parmar N, et al. Attention is all you need. In: Advances in Neural Information Processing Systems (NeurIPS); 2017.

5. Singh S, Kumar M, Kumar A, Verma BK, Abhishek K, Selvarajan S. Efficient pneumonia detection using Vision Transformers on chest X-rays. Sci Rep. 2024;14(1):2487. doi:10.1038/s41598-024-52703-2

6. Aldabayan YS. Pneumonia and pneumothorax detection: a multi-factor evaluation of chest X-rays. PLoS One. 2026;21(1):e0341060. doi:10.1371/journal.pone.0341060